

This is due by August 1.

Problem 1. We have seen PID has the property that every ascending chain of ideals stabilizes. This property turns out to be so important that it has a name, Noetherian (in honor of Emmy Noether).

Definition. Let R be a commutative ring. We say R is Noetherian if every ascending chain of ideals stabilizes. That is, if $I_0 \subseteq I_1 \subseteq I_2 \subseteq \cdots$, there exists some $N \in \mathbb{N}$ such that $I_N = I_{N+1} = \cdots$.

Let R be a commutative Noetherian ring. Let S be a commutative ring.

1. Determine if any field, $\mathbb{Z}$, and $\mathbb{Z}_m$ is Noetherian.
2. There is a dual notion to being Noetherian called Artinian (named after Emil Artin).

Definition. A commutative ring R is Artinian if every descending chain of ideals stabilizes. That is, if $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$, there exists some $N \in \mathbb{N}$ such that $I_N = I_{N+1} = \cdots$.

Determine if $\mathbb{Z}_n$, any field, and $\mathbb{Z}$ is Artinian.

3. Show a commutative ring is Noetherian iff every ideal is finitely generated.

Proof. 1. A field k has only two ideals (0) and k . So any ascending chain of ideals only depends on where k shows up the first time because it must be (0) before that and k after that. It will either stabilize to (0) or (k) .

$\mathbb{Z}$ is a PID so any ascending chain of ideals looks like $(d_0) \subseteq (d_1) \subseteq \cdots$ for $d_i \geq 0$ which is equivalent to $\cdots \mid d_1 \mid d_0$. This implies $\cdots \leq d_1 \leq d_0$. If the chain does not stabilize, then we must have infinitely many strict inequalities. But any strictly descending sequence of nonnegative integers terminates after finitely many terms.

Recall $\mathbb{Z} \rightarrow \mathbb{Z}_m$ is surjective. For any ascending chain of ideals in $\mathbb{Z}_m$, the preimages is an ascending chain of ideals in $\mathbb{Z}$ which must stabilize. Therefore the images must stabilize.

2. A field is Artinian by the same argument in step 1 because any descending chain of ideals is completely determined by the first time (0) shows up.

$\mathbb{Z}$ is not artinian because $(2) \supseteq (4) \supseteq (8) \supseteq \cdots$ is a nonstabilizing chain of ideals.

$\mathbb{Z}_n$ has finitely many elements and hence has finitely many ideals. Then any infinite chain of ideals must repeat. So any descending chain of ideals must stabilize because when it repeats we get $I_n \supseteq I_{n+1} \supseteq \cdots \supseteq I_n \supseteq I_{n+1}$ which implies $I_{n+1} = I_n = \cdots$.

3. ($\implies$) Suppose R is Noetherian. Suppose I is not finitely generated. So say $x_0, x_1, \dots$ are the generator. Then $(x_0) \subseteq (x_0, x_1) \subseteq \cdots$ is an ascending chain of ideals that does not stabilize. ($\impliedby$) Let $I_0 \subseteq I_1 \subseteq I_2 \subseteq \cdots$ be an ascending chain of ideals. We claim $I = \bigcup_{n \geq 0} I_n$ is an ideal. Clearly $0 \in I_0 \subseteq I$. For any $x, y \in I$, say $x \in I_p$ and $y \in I_q$. Then $x, y \in I_{\max\{p, q\}}$ which is an ideal. So $x + y \in I_{\max\{p, q\}} \subseteq I$. Let $r \in R$ and $x \in I$. Then $x \in I_p$ which is an

ideal. It follows $rx \in I_p \subseteq I$. Then I is finitely generated, say with generators $a_0, \dots, a_t$. Then these generators must eventually all show up in the ascending chain of ideals. After that, the ascending chain of ideals stabilizes $\square$

Problem 2. Let R be a commutative ring and x be an indeterminate. Define

$$R[x] = \{a_0 + a_1x + \dots + a_nx^n : a_i \in R \text{ for all integers } 0 \leq i \leq n < \infty\}$$

1. Assume $R[x]$ is a ring. Show if R is a domain, then $R[x]$ is a domain.
2. Show $(2, x)$ is not a principle ideal in $\mathbb{Z}[x]$.
3. Determine if $(x - 3)$ is a prime ideal in $\mathbb{Z}[x]$. (Is it maximal?)

Proof. 1. Suppose $f(x)g(x) = 0$ for some $f(x), g(x) \in R[x]$. Then $LC(f)LC(g) = 0$ in $\mathbb{R}$. Since R is a domain, WLOG, we may assume $LC(f) = 0$ which is impossible.

2. Suppose $(2, x) = (f)$ for some $f \in \mathbb{Z}[x]$. Then $f \mid 2$ which implies $\deg f \leq \deg 2 = 0$. Since $(2, x) \neq (0)$, we get $\deg f = 0$. That is, $f \in \mathbb{Z}$ and we write $f = \pm 1$ or ± 2 . As $(2, x) \neq \mathbb{Z}[x]$, $f = \pm 2$ but $\pm 2 \nmid x$.

3. $(x - 3)$ is irreducible in $\mathbb{Z}[x]$ which is an UFD. So it is prime and generates a prime ideal. It is not maximal because $(x) \subseteq (x - 3, 3) \neq R$. $\square$

Problem 3. Consider $R = \mathbb{Z}_2[x]$ (it is a ring by Problem 2).

1. Show R is a domain and find $U(R)$.
2. Is $x^3 + x^2 + 1$ irreducible in R ? (You may first find all deg 2 irreducible elements.)
3. Let $R' = \mathbb{Z}_4[x]$. Is there a unit element of degree 1? (Contrasts to part 1.)

proof of part 3. Consider $(2x + 1)^2 = 1$ in $\mathbb{Z}_4[x]$.

(Albert.) Note $(2x + 1)(2x + 3) = 3 = -1$.

(Tianshuo.) Note $1 + 2x = 1 - 2x$. So $(1 + 2x)^2 = (1 + 2x)(1 - 2x) = 1$. $\square$

Problem 4. Like group homomorphism, there are special functions connecting rings.

Definition. Let R, S be commutative rings. A function $f : R \rightarrow S$ is a ring homomorphism if for all $a, b \in R$

$$f(1_R) = 1_S, \quad (a + b) = f(a) + f(b), \quad f(ab) = f(a)f(b).$$

Let $f : R \rightarrow S$ be a ring homomorphism. Show the following.

1. Show $\text{im}(f) = \{s \in S : s = f(r) \text{ for some } r \in R\}$ is a subring and $\ker(f) = \{r \in R : f(r) = 0\}$ is an ideal.
2. Let $I \subseteq S$ be an ideal. Show $f^{-1}(I) = \{r \in R : f(r) \in I\}$ is also an ideal.
3. Suppose f is surjective. Let $J \subseteq R$ be an ideal. Then $f(J)$ is an ideal.
4. Give a ring homomorphism $f : R \rightarrow S$ such that $f(J)$ is not an ideal while J is an ideal.
5. (Just Think about it.) Define $\rho : \mathbb{Z} \rightarrow \mathbb{Z}$ by $n \mapsto n + 1$. Note $\rho^{-1}(2\mathbb{Z}) = 2\mathbb{Z} + 1$. Does this example contradicts the result in part 2?

proof of part 4. By part 3, we must consider a nonsurjective homomorphism. For instance, the inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$ does not take $2\mathbb{Z}$ to an ideal in $\mathbb{Q}$.

(Marcial.) The inclusion $\mathbb{Z} \rightarrow \mathbb{Z}[x]$ does not take $2\mathbb{Z}$ to an ideal in $\mathbb{Z}[x]$. □

Proof. (Nikoli.) ρ is not a ring homomorphism. □